



T(6th Sm.)-Computer Sc.-G/(DSE-B-3)/CBCS

2021

COMPUTER SCIENCE — GENERAL

Paper : DSE-B-3

(Computational Mathematics)

Full Marks : 50

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **question no. 1** and **any four** from the rest.

1. Answer **any five** questions :

2×5

- (a) What do you mean by planar graph? Give example.
- (b) Suppose there are two simple graphs G_1 and G_2 . How do you verify whether G_1 and G_2 are isomorphic?
- (c) Define path and circuit.
- (d) Name two direct methods to solve a system of linear equations.
- (e) What are the limitations of Newton–Raphson method?
- (f) State the condition for convergence of an iteration method.
- (g) What is the condition for convergence of Gauss-Seidel method of iteration?
- (h) State the drawback of Simpson's $1/3$ rd rule for solving a definite integral.

2. (a) Find $f(1895)$ using Newton's Forward Difference formula

x	1891	1901	1911	1921	1931
$f(x)$	46	66	81	93	101

(b) State Newton–Raphson formula and criteria for convergence.

4+(3+3)

3. (a) What is Absolute error and Relative error? Explain with an example.

(b) Solve the given equations using Gauss-Jordan method :

$$x + 2y + 6z = 66$$

$$3x + 4y + z = 78$$

$$6x - y - z = 57$$

(3+3)+4

4. (a) Prove that the sum of the degrees of the vertices of any finite graph is even.

(b) A simple graph G has 24 edges and degree of each vertex is 4. Find the number of vertices. Write down the formula used.

5+5

Please Turn Over

5. (a) Prove that the rate of convergence of Bisection method is linear.
 (b) Solve by Gauss elimination method, the following system of equations:

$$2x - y + 3z = 9$$

$$x + y + z = 6$$

$$x - y + z = 2$$

5+5

6. (a) Define Euler graph and Hamiltonian path with proper examples.
 (b) Proof that a simple graph with n vertices and k components can have at most $(n-k)(n-k+1)/2$ edges. (3+3)+4
7. (a) What is the difference between Newton–Raphson and Regula Falsi method? Discuss.
 (b) Consider the following dataset :

Temp. in Celsius	140	150	160	170	180
Pressure	3.685	4.854	6.302	8.076	10.225

Find the pressure at 175 degree Celsius using Newton's backward interpolation.

5+5

8. (a) Find a root of an equation $x^3 - x - 1 = 0$ using Secant method correct up to two decimal places.
 (b) Find the root of the equation $2x^3 - 2x - 5 = 0$ using Bisection method correct up to 3 places of decimal. 5+5
-